

range is calculated, which is the crow-flight distance.

Typically, a TACAN transponder of a carrier can provide navigational data to approximately 100 aircraft, simultaneously interrogating at a time. With TACAN, an aircraft can accurately navigate to the carrier. Direction and range are displayed in the aircraft's moving map display as waypoints and also as pointers in the pilot's head-up display (HUD). (Refer 'Fighter Aircrafts,' article published in July 2007, August 2007 and September 2007 issues.)

The next task is to control the movements of outbound and inbound aircraft. Appropriate commands and support are given to inbound aircraft for landing. This is done by carrier air-traffic control, which is generally called amphibious air-traffic control.

Air-traffic control

Since both the carrier and aircraft are moving, air-traffic control is much more complex and elaborate than their land based counterparts. All these operations are controlled from a command centre called carrier air-traffic control centre (CATCC). This command centre is situated on the top-most storey, called Pri-fly, above the flight-deck of a carrier. All aviation-related data like aircraft's flight plans and approaching aircraft's status are assembled at this centre for air operations to be monitored and controlled.

For these control operations, many radar systems play a crucial role. Outbound aircraft are allotted a 3D air corridor through which these leave without interfering with incoming aircraft. Incoming aircraft from various locations are first marshalled to another air corridor and then assigned height, direction and speed to fly in a pattern around the CSG.

Depending on the status of the aircraft—whether low on fuel or hit by enemy but in a flyable condition or having technical snag—these are assigned priorities. As incoming aircraft of high priority keep land-



Fig. 24: Antenna of AN/SPN-43 marshalling radar

ing, aircraft flying around the CSG are instructed to move to hold pattern around the carrier. As aircraft keep landing, a landing sequence is formed where aircraft trail behind each other by kilometres.

Once aircraft are lined for landing, these are transferred to approach and landing controllers, that help the pilots in landing the aircraft. During high-intensity air operations aircraft come and land every 30 seconds. For this, a radar called marshalling radar is used, which is the air-traffic control radar.

Marshalling radar. The marshalling radar AN/SPN-43 is a 2D air-surveillance S-Band (2GHz-4GHz) radar, dedicated for air-traffic control. It is used for vectoring carrier-bound aircraft into final approach for landing. It also acts as a back-up, short-range, air-search radar for air-defence operations. It provides azimuth and range of detected aircraft from 80.5km (50 miles) to a minimum range of 229m (250 yards), at altitudes from radar horizon to 9144m (30,000 feet).

The radar has an inbuilt Identify Friend or Foe (IFF) system. The radar display console displays all echoes in the area of responsibility in the form of blips. But how can individual aircraft be set apart from these blips? For this, a system called Direct Altitude and Identity Readout (DAIR) is used to identify the aircraft.

DAIR. DAIR is a highly-automated beacon and radar-tracking system.



ENVIRONMENT TESTING INSTRUMENTS



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920P

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Range	0 ~ 200,000 lux, 0 ~ 20,000 fc

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Range	0% ~ 100% RH
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